Narainkarthigeyan Singaram Balamurugan

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Education

Georgia Institute of Technology May 2028
B.S. in Computer Science, Concentration: Artificial Intelligence and People *GPA: 3.89/4.0*

- **Relevant Coursework:** Data Structures and Algorithms, Objects and Design, Introduction to Artificial Intelligence, Linear Algebra, Computer Organization and Programming, Object-Oriented Programming
- **Activities/Honors:** Dean's List, Faculty Honors, Trading @ GT, Data Science at GT, Startup Exchange

Experience

Software Engineer Intern May 2026 – August 2026
Magnitudo Technologies *Remote*

- Engineered a **full-stack student analytics platform** for academic, attendance, and extracurricular data by building **RESTful microservices** in **Next.js 16, React 19, and Node.js**, enabling real-time administrative oversight across multi-class cohorts.
- Built an automated risk-scoring microservice in **Python/FastAPI** using weighted heuristics for GPA decay, attendance thresholds, and disciplinary NLP flags, reducing manual review workload and **flagging 50+ at-risk students**.

Machine Learning Researcher January 2026 – May 2026
Georgia Tech VIP Program *Atlanta, GA*

- **Reduced an autonomous-vehicle speed controller's optimization cost from 473,501.9 to 2.55** by implementing a **Python objective function** that penalized unsafe acceleration, speed deviation, red-light violations, & obstacle collisions.
- **Improved LLM decision consistency** by testing multiple prompt formats, enforcing **strict JSON outputs**, and **lowering inference temperature from 0.7 to 0.2** for more stable vehicle-control responses.

AI Software Engineer Intern June 2024 – August 2024
VDart *Johns Creek, GA*

- Engineered a production-grade multimodal RAG chatbot using **Python & LangChain**, achieving **98% accuracy across 3.5K+ queries**; benchmarked **5+ LLMs** to automate workflows and **reduce support load by 40%**.
- Architected a robust **SQL** validation suite for **250+ queries** using **SQL Workbench & Matplotlib**, **improving data reliability by 90%** through automated anomaly detection and visual reporting.

Projects

FlashSeat — High-Concurrency Reservation Engine | *Go, Next.js, TypeScript, PostgreSQL, NoSQL, Redis, Docker, REST APIs* Aug. 2026

- Architected a real-time reservation engine in **Go, Next.js, and TypeScript** that prevented double-bookings across **1,000+ concurrent HTTP clients** by combining atomic **Redis Lua scripts** with **PostgreSQL** row locks.
- Achieved **64 ms P99** reservation latency and a 100% pass rate on concurrency invariants across **1,700 real HTTP requests** by building a **Dockerized k6 load-testing** and correctness-validation pipeline.

Binance L2 Order-Book Engine & Synthetic L3 Queue Simulator | *C++20, WebSockets, mmap* July. 2026

- Engineered a zero-allocation **C++20 L2 order-book** replay path using fixed-capacity Robin Hood hash tables, indexed heaps, fixed-point arithmetic, and memory-mapped I/O, **sustaining 7.08M price-level updates/sec**.
- Reconstructed synthetic **Level-3 order queues** from Level-2 market-by-price feeds by developing a **256-particle Monte Carlo simulation** that tracks FIFO queue priority and calculates fill probability quantiles **(P10/P50/P90)**.

Melanoma Detection AI App | *Python, PyTorch, CNNs, Grad-CAM, OpenCV, CUDA* Mar. 2026

- Developed an end-to-end **PyTorch computer-vision pipeline** for seven-class skin-lesion classification on the HAM10000 dataset using a **custom CNN**, EfficientNet-B0 transfer learning & conditional DCGAN augmentation.
- Increased test accuracy from **44.9% to 77.6%** and more than doubled macro F1 from **0.33 to 0.68** using **CUDA-accelerated training** and **OpenCV** image preprocessing.

Polaris.ai — Multi-Agent Job Intelligence Platform | *React.js, Node.js, LangGraph, LangChain, Firebase* Dec. 2025

- Architected a hierarchical 6-agent **LangGraph** framework in **Node.js/Express** to execute autonomous candidate matching across **2,500+ live listings**, cutting median time-to-application by **95%**
- Engineered an AI-powered **ATS optimization** engine via typed **GraphQL mutations**, handling binary resume uploads to compute real-time keyword deltas and driving a increase in candidate alignment scores.

Technical Skills

Languages: Python, JavaScript, TypeScript, Java, C, C++, Go (Golang), SQL, HTML, CSS, MATLAB
Frameworks & Libraries: React.js, Next.js, Node.js, Vue.js, Django, Flask, PyTorch, TensorFlow, OpenCV, Pandas, LangGraph, Matplotlib
Cloud Tools & Databases: AWS (EC2, S3, Lambda), Google Cloud Platform (GCP), Docker, Kubernetes, CI/CD, GitHub Actions, Linux, Git, PostgreSQL, MongoDB, Firebase, GraphQL, REST APIs